

Optimal Execution Model

Market Microstructure Analysis & Optimal Liquidation Trajectory · Gianluca Pogliana - +39 3403276133 - poglianagianluca@gmail.com

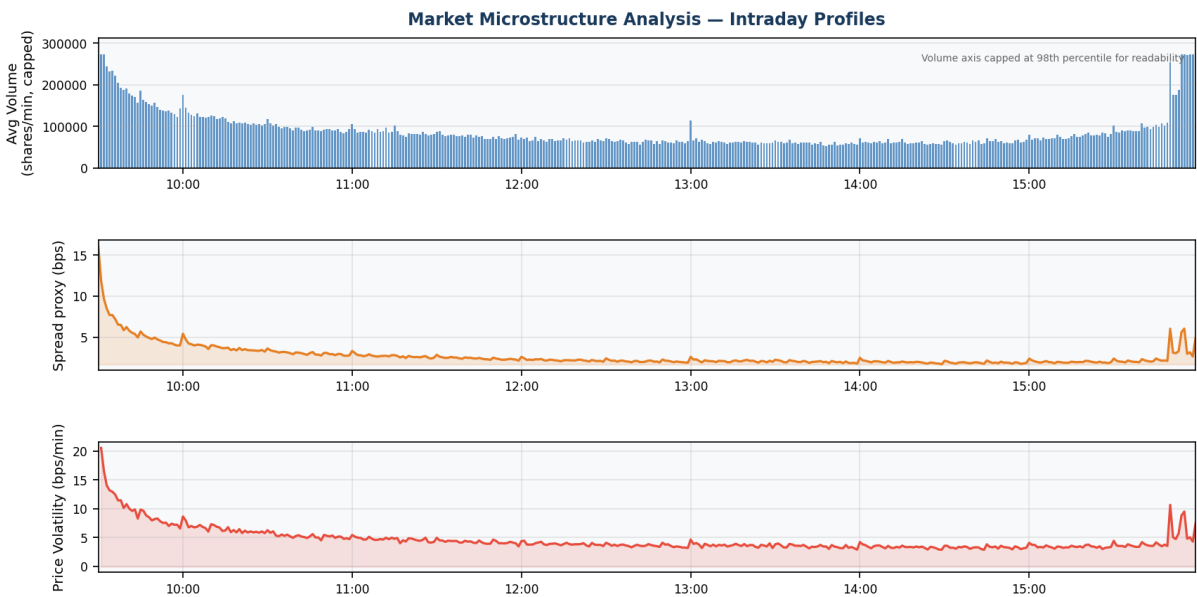
1. Executive Summary

This project develops a quantitative execution framework for a large **AAPL** liquidation order. The objective is to compare standard benchmark execution schedules, such as TWAP and VWAP, with an **Almgren-Chriss optimal trajectory** calibrated on real intraday market data. The model is designed to answer a practical execution-desk question: how should a large parent order be split over the trading day in order to balance expected transaction costs against the risk of adverse price movements?

Project Snapshot	
Instrument	AAPL equity 1-minute bars, regular session (09:30-16:00 ET) 2024-10-14 – 2026-10-08 (498 trading days)
Parent Order	100,000 shares Notional ~USD 34,042,000
Execution Setup	1.0 trading day 100 execution slices TWAP, VWAP and Almgren-Chriss comparison

2. Market Microstructure Analysis

2.1 Intraday Profiles — AAPL (Polygon API)



Metric	Value	Metric	Value
Peak Volume Time	09:30	Avg Spread Proxy	2.7 bps
Min Spread Time	14:29	Max Spread Proxy	16.2 bps
ADV	36,749,229 shares	Last Price	USD 340.42

2.2 Interpretation & Execution Relevance

Peak volume occurs at **09:30** (market open), confirming the U-shaped intraday volume pattern. The range-based spread proxy (not a quoted bid-ask spread) reaches a maximum of **16.2 bps** at the open and compresses to a daily minimum at **14:29**, with an intraday average of **2.7 bps**. These findings suggest that the most favourable execution window for minimising direct transaction costs lies between **10:30 and 14:00**, when spreads are lower and intraday volatility is more stable. Volume is highest at open and close and lowest at midday. High-volume windows offer more

liquidity per unit of time, but they coincide with wider spreads and higher volatility, so the trade-off between impact, spread and timing risk is what the model in Section 3 quantifies. The spread is widest at open, when uncertainty and adverse selection risk are highest, and tightens during the day as liquidity improves. Intraday volatility is highest at open and close, increasing timing risk for patient execution strategies.

3. Almgren-Chriss Optimal Execution Model

3.1 Problem Statement & Closed-Form Solution

Liquidating a large order exposes the trader to a fundamental trade-off: **slow execution** reduces market impact but increases timing risk (price drift); **fast execution** reduces timing risk but amplifies impact costs. Almgren & Chriss (2001) solve this formally by minimising:

$$\min E[\text{Cost}] + \lambda \times \text{Var}[\text{Cost}]$$

The closed-form optimal trajectory is $x(t) = X \times \sinh(\kappa(T-t)) / \sinh(\kappa T)$ where $\kappa = \sqrt{(\lambda\sigma^2/\eta)}$ is the urgency parameter. For $\lambda = 2.46\text{e-}08$ (set so that the first slice is 1.8x the TWAP slice), $\kappa = 1.7011$: the strategy is moderately front-loaded, executing 1.8% in the first slice vs 1.0% for a uniform TWAP.

Var	Value	Description
X	100,000 shares	Total order size
T	1.0 day	Execution horizon
N	100 slices	Execution intervals
λ	2.46e-08	Risk aversion (set so first slice = 1.8x TWAP slice)
κ	1.7011	Urgency parameter $\sqrt{(\lambda\sigma^2/\eta)}$
σ	1.805%/day	Daily price volatility
η	3.21e-07 USD/share ²	Temporary impact coefficient
γ	3.21e-08 USD/share ²	Permanent impact coefficient
ADV	36,749,229	Average daily volume (shares)

3.2 Execution Trajectories

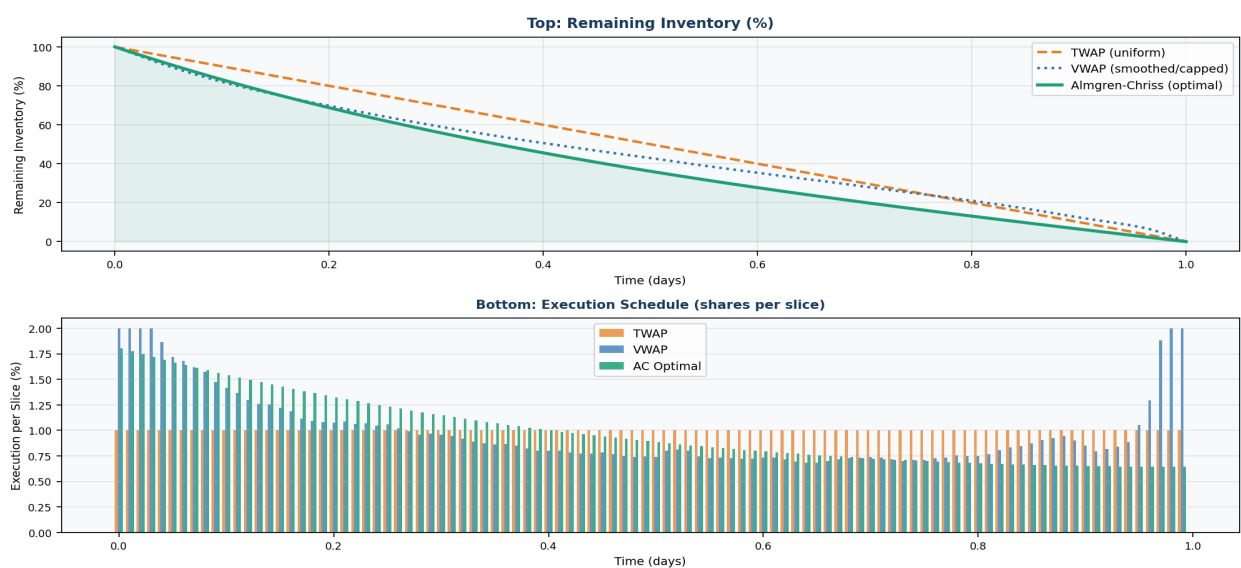


Fig. 2 — Top: remaining inventory (%) for TWAP, VWAP and AC Optimal over the trading horizon. Bottom: shares executed per slice (%). AC front-loads execution vs TWAP (first slice: 1.8% vs 1.0%) to reduce timing risk; VWAP concentrates at open/close following the volume profile.

The AC Optimal strategy has a **modelled expected execution cost** of **USD 3,735** (+11.0% vs TWAP), achieving a lower-risk trajectory by front-loading execution while containing market impact. VWAP has a higher modelled expected cost (**USD 3,798**, +12.9% vs TWAP) because its smoothed volume-based schedule still allocates larger slices around the open and close. Since the model penalises trading intensity through a squared trade-size term, more concentrated execution increases expected temporary impact relative to a uniform TWAP schedule. In practice, VWAP remains widely used because it minimises *deviation from the market average price*, not necessarily because it minimises model-implied implementation shortfall.

The cost figures reported below are **model-implied expected execution costs**, not realised trading costs. They are derived from the Almgren-Chriss impact function and should be interpreted as an implementation-shortfall proxy under the model assumptions. They do not include actual broker commissions, exchange fees, realised bid-ask spread paid, taxes or slippage from real fills. Their main purpose is to compare execution trajectories on a consistent basis.

Strategy	Modelled Cost	vs TWAP	Execution Logic	Best Use Case
TWAP	USD 3,365 (1.0 bps)	—	Uniform execution across all slices	Stable, liquid and low-volatility markets
VWAP	USD 3,798 (1.1 bps)	+12.9%	Execution follows the observed volume profile	Benchmark-sensitive execution versus market VWAP
AC Optimal	USD 3,735 (1.1 bps)	+11.0%	Risk-adjusted schedule minimising $E[\text{Cost}] + \lambda \text{Var}[\text{Cost}]$	Institutional block execution with explicit risk-cost trade-off

The strategy comparison highlights the central trade-off faced by an execution desk: minimising expected transaction costs versus reducing exposure to adverse price movements during the execution horizon.

TWAP achieves the lowest modelled expected cost (**USD 3,365**) because it distributes the parent order uniformly across the trading day. However, its simplicity comes at the cost of higher timing risk: a large portion of the order remains exposed to market movements until late in the execution window.

VWAP follows the historical intraday volume profile and is useful when the trader is benchmarked against market VWAP. In this case, however, VWAP produces a higher modelled cost (**USD 3,798**, +12.9% vs TWAP) because its schedule trades larger slices around the open and close, where AAPL volume is concentrated; with a constant temporary-impact coefficient, the squared trade-size term makes uneven schedules more expensive than a uniform one.

Almgren-Chriss provides the most balanced institutional execution trajectory. The calibrated schedule executes **1.8%** in the first slice versus **1.0%** for TWAP, reducing inventory exposure while avoiding the excessive market impact of immediate execution. The resulting cost is higher than TWAP but reflects a more explicit risk-adjusted execution objective.

3.3 Efficient Frontier

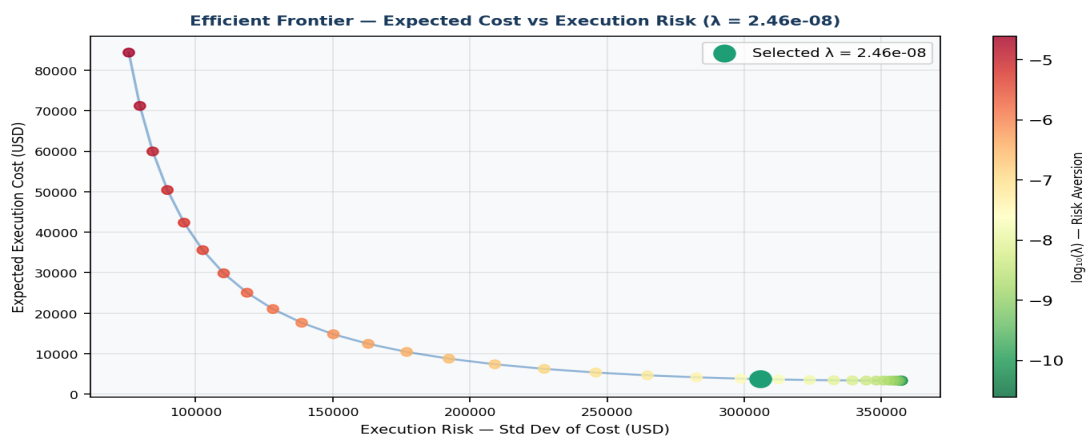


Fig. 3 — Efficient frontier: expected execution cost versus execution risk for varying λ . Moving left along the frontier corresponds to more aggressive execution: risk decreases, but expected cost increases due to stronger market impact. Moving right corresponds to more patient execution: expected cost decreases, but timing risk increases. Green dot = selected $\lambda = 2.46e-08$.

The frontier illustrates the irreducible trade-off between execution cost and execution risk. At low λ the strategy is patient (TWAP-like), accepting high timing risk to minimise market impact. At high λ it becomes aggressive, front-loading execution to eliminate price risk at the cost of higher impact. The selected $\lambda = 2.46\text{e-}08$ positions the strategy at the patient end of the frontier, consistent with a mildly front-loaded schedule.

3.4 Strategy Selection Matrix

Strategy	Strengths	Weaknesses	Typical Use
TWAP	Simple, transparent and operationally easy to implement	Ignores liquidity profile and leaves high residual inventory risk	Passive execution in stable and liquid markets
VWAP	Tracks the market volume curve and aligns with a common benchmark	Can trade too aggressively during high-spread or high-volatility windows	Benchmark-driven execution for asset managers
Almgren-Chriss	Explicitly balances expected market impact and timing risk	Requires calibration of impact and risk-aversion parameters	Institutional block execution

4. Price Impact Estimation

Price impact is estimated using the **square-root market impact model** (Almgren et al., 2005): $MI = Y \times \sigma_{\text{daily}} \times \sqrt{(Q/ADV)}$, with $Y = 0.1$ (an assumption, not calibrated on fills). This provides an indicative pre-trade benchmark for assessing the size of the order relative to market liquidity. For an order of **100,000 shares** against an ADV of **36,749,229 shares**, the participation rate is **0.3%** of daily volume. The estimated total impact benchmark is **USD 4,808** (**1 bps** of notional), providing a reference point for evaluating the scale of the execution problem.

Parameter	Value	Formula / Note
Daily Volatility (σ)	1.80%	Std(daily returns)
Annualised Volatility	28.7% p.a.	$\sigma_{\text{daily}} \times \sqrt{252}$
Avg Daily Volume (ADV)	36,749,229 shares	Mean of daily volume
Participation Rate	0.3%	Q / ADV
Last Price	USD 340.42	Most recent close
Temp. Market Impact	USD 0.0321/share	$Y \times \sigma_{\text{daily}} \times \sqrt{(Q/ADV)} \times P$
Perm. Market Impact	USD 0.0160/share	50% of temporary impact
Total Impact Cost	USD 4,808	(1 bps of notional)

Data and assumptions. Volume and prices are 1-minute aggregate bars from Polygon (regular session only). The spread series is a heuristic range-based proxy, not NBBO quotes. The impact coefficient $Y = 0.1$ is an assumption; the AC temporary-impact coefficient η is derived from the same square-root benchmark ($\eta = Y \cdot \sigma \cdot P \cdot \sqrt{(Q/ADV)/Q}$), so Sections 3 and 4 are consistent. λ is not estimated from a utility function: it is set to obtain a chosen first-slice size. ADV comes from Polygon aggregate bars and may differ from the consolidated volume shown by other providers, so absolute figures are indicative while the comparison between schedules is unaffected. All costs are model-implied and are meant to compare schedules, not to forecast realised costs.

5. Conclusions

This project demonstrates how optimal execution theory can be combined with real market microstructure information to design institutional execution strategies. By integrating intraday volume, a range-based spread proxy, realised volatility and market impact estimation, the framework provides a practical tool for comparing benchmark execution algorithms against a risk-adjusted Almgren-Chriss trajectory.

For the analysed AAPL order, the calibrated Almgren-Chriss strategy deliberately accepts a moderate increase in expected cost versus TWAP (+11.0%) in exchange for a more controlled inventory profile and lower exposure to adverse price movements. This is consistent with the way an execution desk evaluates large orders: not only by expected cost, but by the balance between market impact, timing risk and benchmark sensitivity.

Although simplified relative to production execution systems, the model captures the fundamental economic trade-off underlying modern electronic execution. Natural extensions include stochastic liquidity, adaptive participation rates, order-book imbalance signals, limit-order execution and reinforcement-learning based execution policies.

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